EXPERIMENTATION AND DECISION-MAKING WITH WQU APPLICANT DATA

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1. EXECUTIVE SUMMARY & OVERVIEW

This technical report presents a quantitative evaluation of applicant data, experimental design framework, and unsupervised customer/applicant segmentation models completed under the WorldQuant University (WQU) Data Science Lab framework. The primary objective is to evaluate applicant engagement, optimize decision-making workflows via rigorous statistical hypothesis testing, and perform dimensionality reduction alongside clustering to segment populations effectively.

2. EXPERIMENTAL DESIGN & HYPOTHESIS TESTING

To measure the impact of interventions on applicant completion rates, an A/B testing methodology is established. We define control (A) and variant (B) cohorts across applicant demographics. The metric of interest is the applicant conversion rate (completion of assessment modules).

2.1 Statistical Formulation

Let p_A and p_B represent the true underlying completion probabilities for the control and variant groups, respectively. The null hypothesis (H_0) and alternative hypothesis (H_1) are formulated as:

$$H_0: p_B - p_A \leq 0 \quad \text{vs.} \quad H_1: p_B - p_A > 0$$

The test statistic for the two-sample z-test for proportions is computed via pooled standard error:

$$Z = (\hat{p}_B - \hat{p}_A) / \sqrt{[\,\hat{p}(1-\hat{p})\,(1/n_A + 1/n_B)\,]}$$

where $\hat{p} = (x_A + x_B) / (n_A + n_B)$ represents the pooled proportion, with x denoting successes and n denoting total group sizes.

3. DEMOGRAPHIC & BEHAVIORAL DATA SUMMARY

COHORT ID	SAMPLE SIZE (N)	COMPLETION RATE (%)	MEAN ASSESSMENT SCORE	STANDARD ERROR
Control Group (A)	12,450	42.3%	74.2 / 100	0.0044
Variant Group (B)	12,510	47.8%	78.6 / 100	0.0045
Pooled Population	24,960	45.05%	76.4 / 100	0.0031

4. DIMENSIONALITY REDUCTION & UNSUPERVISED SEGMENTATION

High-dimensional applicant features are compressed using Principal Component Analysis (PCA) prior to unsupervised clustering via the k-Means algorithm. Feature scaling via standard Z-score normalization precedes covariance decomposition.

4.1 Principal Component Analysis (PCA)

Eigenvalues λ_i and eigenvectors $\mathbf{v}_i$ of the feature covariance matrix $\mathbf{\Sigma}$ satisfy:

$$\mathbf{\Sigma}\,\mathbf{v}_i = \lambda_i\,\mathbf{v}_i$$

The cumulative explained variance ratio for k components is defined by:

$$\text{Cumulative Variance Ratio} = \left(\sum_{i=1}^{k}\lambda_i\right) \Big/ \left(\sum_{j=1}^{p}\lambda_j\right)$$

5. SEGMENTATION MODEL PERFORMANCE

CLUSTER COUNT (K)	INERTIA (WCSS)	SILHOUETTE SCORE	CALINSKI-HARABASZ INDEX
k = 2	45,210.4	0.312	1,240.5
k = 3	31,840.1	0.428	1,890.2
k = 4	24,105.8	0.389	1,650.8
k = 5	19,870.3	0.341	1,420.1

6. KEY FINDINGS & STRATEGIC DECISION FRAMEWORK

- **A/B Testing Outcome:** The experimental intervention yielded a statistically significant increase in completion rates ($Z = 8.74$, $p < 0.0001$), rejecting the null hypothesis.
- **Optimal Segmentation:** Evaluation of Silhouette scores and Calinski-Harabasz metrics confirms $k = 3$ as the optimal structural partitioning for applicant profiles.
- **Decision Implementation:** Operational workflows should tailor onboarding communications specifically toward high-potential candidate clusters identified through principal component projection.